ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN

Submarine Naval Communicator (00299)

Qualification Code: (AIIW)

MITE IDs

109270 AIIW
109271 (OJT Annex E)

RECORD OF AMENDMENTS

The "Authority" reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of the Naval Staff (C Naval). Approval responsibility for the QSP has been granted to Directorate Maritime Training & Education (DMTE). Approval for changes to chapters 3 and 4 may be further delegated to the Commandant Canadian Forces Naval Operation School (CFNOS). The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J "Course History".

REVISION	APPROVAL DATE	AUTHORITY
1	10 December 2004	Cdr E.P. Deslauriers, DMTE
2	11 FEB 2013	Cdr R.J. Watt, DMTE
2.1	02 FEB 2021	NTDC(A)

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FOREWORD

1. This training document is developed under the authority of the C Naval and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055-003/PQ-001 Occupation Submarine Naval Communicator Specialty Specification (NCM) dated 2001/05/08. Qualification Code AIIW.

2. This publication is effective on receipt. Performance Objectives (PO) are explained fully in Chapter 2. The Lesson Specifications (LS)/Enabling Objectives (EO), Enabling Checks (EC) and Performance Checks (PC) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). QSP revision procedures are outlined in the NTI. Amendments to this QSP may only be approved by the Commandant of CFNOS. Authority to approve amendments that affect chapters 3 and 4 may be delegated to the Standards Division Commander.

4. The designated CCA(s) for this training is indicated below:

CCA	PHASE	QUALIFICATION
CFNOS	In School Course	109270
CFNOS	OJT Annex E of QSP	109271/AIIW

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY DMTE

AIM

1. The aim of this qualification standard and plan (QSP) is to provide the Submarine Naval Communicator with the Skills and Knowledge required performing duties on board HMC Submarines and for further training within the Submarine.

2. This QSP is designed to assist instructors in preparing and organizing themselves for the Submarine Naval Communicator Course and should be read prior to commencing any preparation or scheduling. Any suggestions for the improvement of this document should be brought to the attention of the standards division of the Curriculum Control Authority.

COURSE DURATION CALCULATION

3. Course duration is calculated, in accordance with MARCORD 9-30.

Minimum (2 students)

Actual Training Days (ATT)	19
Training Administrative Time (TAT) (if required)	2
Base Administrative Time	1
Course Duration (total calendar days)	22

Optimal (4 students)

Actual Training Days (ATT)	23
Training Administrative Time (TAT) (if required)	2
Base Administrative Time	1
Course Duration (total calendar days)	26

Maximum (6 students)

Actual Training Days (ATT)	31
Training Administrative Time (TAT) (if required)	3
Base Administrative Time	1
Course Duration (total calendar days)	35

RESOURCES

4. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H.

TITLE	RANK	QUALIFICATION	QTY	REMARKS
Course Officer	LT(N)	Mars 71C	1	
Divisional CPO	CP02/PO1	ANY	1	
Instructor	PO1/PO2/MS	AILS, AIIW & QL6A Qualified	1	

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Communication Trainer or Submarine alongside	Min/Opt/Max 15 / 23 / 35

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

a. Maximum – 6

b. Minimum – 2

c. Optimum – 4

COURSE PREREQUISITES

6. The candidates must:

a. Be Naval Communicator (00299) QL5A qualified (AIXC); and

b. Should have completed Basic Submarine Qualification (AILS) training.

RELATED DOCUMENTS

7. The following documents are related to this QSP:

a. CISN OP ID: 114949; and

b. Basic Submarine Qualification.

EVALUATION

8. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques.

9. Trainee evaluation shall be conducted in accordance with NTI 01/2008 and chapter 3 of this QSP. In order to pass this course, learners must successfully complete all POs.

VALIDATION

10. This QSP is used for the preparation of validation surveys. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

11. Successful completion of this course and OJPR constitutes the primary means of attaining the Submarine Naval Communicator Qualification, Code AIIW - MITE codes 109270 and 109271.

PRIOR LEARNING ASSESSMENT

12. Prior Learning Assessment shall be conducted in accordance with A-P9-050-000/PT012 Canadian Forces Military Equivalencies Program and MARCORD 9-47.

SPECIAL INSTRUCTIONS

13. Nil

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY DMTE

PO 001 CONFIGURE MESSAGE HANDLING DISTRIBUTION SYSTEM (MHDS)

1. Performance: Configure Submarine Message Handling Distribution System (MHDS)

2. Conditions:

   a. Given:

      (1) MHDS terminal; and

      (2) References.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Submarine at sea or alongside.

3. Standard. The Radio Watch Keeper shall, IAW references configure Submarine Message Handling And Distribution System (MHDS) by:

   a. Configuring MHDS for submarine broadcast;

   b. Configuring MHDS for submarine ship/shore;

   c. Configuring MHDS for submarine inter/ship; and

   d. Configuring MHDS for submarine system backup.

4. References.

   C1, C2

5. Specification Task Numbers.

   NT0006, NT0007

6. Training Limitations.

   If trainer not available use of submarine alongside.

PO 002 OPERATE SUBMARINE COMMUNICATIONS EQUIPMENT

1. Performance: Operate Submarine Communications Equipment.

2. Conditions:

   a. Given:

      (1) Radio Equipment; (2) Operational Keymat; and

      (3) References.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Submarine at sea or alongside.

3. Standard. The Radio Watch Keeper shall, IAW refs Operate Submarine Communications Equipment by:

   a. Conducting broadcast procedures to include; (1) Analyzing traffic lists; and

      (2) Configuring Submarine Communications Equipments by:

         (a) Configuring HF Broadcast;

         (b) Configuring LF Broadcast;

         (c) Configuring Satcom Broadcast;

         (d) Describing Submarine Procedural Messages and Operator-to-Operator (OTO) Messages.

b. Conducting ship/shore procedures to include the configuration of submarine communications equipment by:

   1) Tuning HF transmitters for Submarine Ship/Shore; 2) Configuring DAMA for

   Submarine Ship/Shore; and 3) Configuring RSS answering.

c. Configure submarine communication equipment procedures to include:

   a. Configuring HF Secure Voice;

   b. Configuring DAMA;

   c. Configuring HF Plain Voice;

   d. Configuring DAMA for Secure Satellite Voice;

   e. Configuring UHF for LOS Secure Voice;

   f. Configuring Expendable Communications Buoys (ECB'S);

   g. Configuring UHF for LOS Plain Voice;

   h. Configuring VHF;

   i. Configuring HF CRATT; and

   j. Configuring UHF LOS CRATT.

4. References.

A3, A4, B3, B4, B5, B6, B7, B9, B10, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, B52, C1, C3, C4, C5

5. Specification Task Numbers.

T0001, T0002, T0003

6. Training Limitations.

If no trainer available use of Submarine at sea or alongside; and Submarine must be safe to transmit.

PO 003 PERFORM SUBMARINE COMMUNICATION CENTRE FUNCTIONS

1. Performance: Perform Submarine Communication Centre Functions

2. Conditions:

   a. Given:

      (1) References;

      (2) Messages;

      (3) Message Drafting Software; and

      (4) State Boards.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Submarine at sea or alongside.

3. Standard. The Radio Watch Keeper shall, IAW references, perform Submarine Communications Centre functions by:

   a. Disseminating key information to include:

      (1) Extracting information from submarine formatted messages; and

      (2) Summarizing key information.

   b. Drafting submarine formatted messages; and

   c. Maintaining State Boards.

4. References.

   A3, B12, B13, B15, B9, B10, B11

5. Specification Task Numbers.

   T0004

6. Training Limitations.

   If trainer not available use of submarine alongside.

PO 004 ADVISE COMMAND ON STATUS OF SUBMARINE COMMUNICATIONS

1. Performance: Advise Command on Status of Submarine Communications

2. Conditions:

   a. Given:

      (1) State Boards;

      (2) Logs;

      (3) References;

      (4) Submarine Radio Equipment; and

      (5) Pre-sail checklist.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Submarine at sea or alongside.

3. Standard. The Radio Watch Keeper shall, IAW references, advise Command on Status of Communications by:

   a. Formulating recommendations to Command to include:

      (1) Obtaining Command Priorities by: (a) Analyzing equipment; and (b) Recognizing deficiencies.

   b. Deliver reports to include:

      (1) Delivering written reports; and

      (2) Delivering oral reports.

4. References.

   A3, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, C1, C2, C3, C4

5. Specification Task Numbers.

   NT0008

6. Training Limitations.

   If trainer not available use of submarine alongside.

PO 005 CONDUCT SUBMARINE ATMOSPHERIC MONITORING

1. Performance: Conduct Submarine Atmospheric Monitoring

2. Conditions:

   a. Given:

      (1) Analox;

      (2) Draeger kits and tubes;

      (3) Atmosphere Log;

      (4) Atmospheric State Boards; and

      (5) References (Atmospheric standards).

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Submarine at sea or alongside.

3. Standard. The Radio Watch Keeper shall, IAW references, conduct Submarine specific atmospheric monitoring by:

   a. Utilizing atmospheric monitoring equipment;

   b. Disseminating atmospheric results to include: (1) Updating state boards; and

      (2) Updating logs.

   c. Advising Command on atmospheric results.

4. References.

   B53, B54

5. Specification Task Numbers.

   T0005

6. Training Limitations.

   If trainer not available use of submarine at sea/alongside

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE CFNOS COMMANDANT. APPROVAL AUTHORITY MAY BE DELEGATED TO THE STANDARDS DIVISION COMMANDER FOR THIS CHAPTER.

GENERAL

1. This is a pass/fail course. All Performance Checks (PC) are based on Performance Objectives (PO) in Chapter 2 of this QSP and must be successfully completed to achieve this qualification.

PERFORMANCE CHECKS

2. Summative evaluation for this course is in the form of PCs conducted at the termination training for each PO. As per MARCORDS 9-60, PCs are tests to assess the accomplishment of the performance under the conditions and to the standards stated in the PO. While the majority of the assessments are practical, other forms of evaluation are conducted where practical assessment is not feasible or possible.

3. Conduct of any PC will be IAW this QSP and amplified by CFNOS SO 307, and any other applicable CFNOS SO.

4. Supplemental PCs must be in the same format and have the same standard as the original test. PCs may contain more than one part to facilitate different methods of assessment or to modularize particularly extensive performance requirements. Unless otherwise stipulated, failure of one part of a PC only requires retest on that part and does not require the trainee to be retested on the entire PC.

RESPONSIBILITY FOR TESTING

5. Standards cell provides a measure of quality control by confirming that POs are achieved according to the standards delineated in the QSP. They are responsible for the design, administration and scoring of all Performance Checks. Enabling Checks will be the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Student Progress monitoring including unsatisfactory progress IAW CFNOS SO 307.

TRAINING REVIEW BOARD

7. Training Review Board requirements are contained in MARCORD 9-60 Volume 1 and IAW CFNOS SO 307.

8. Course removal for medical, compassionate or other reasons will be dealt IAW CFNOS SSO 307 regarding TRBs

COURSE REPORTS

9. A CF 377 Canadian Forces Course Report will be prepared IAW DAOD 5031-9. The narrative in Block 17 will indicate the trainee's strengths and weaknesses in relation to specific POs and provide information of use to the trainee's Commanding Officer and Career Manager. Training limitations or incomplete training must be identified on the course report.

STUDENT EVALUATION

LAST DATE AMENDED: 2013-02-14

9. The Trainee will be individually evaluated utilising Enabling Checks and Performance Checks IAW Annex K Evaluation Matrix

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE CFNOS COMMANDANT. APPROVAL AUTHORITY MAY BE DELEGATED TO THE STANDARDS DIVISION COMMANDER FOR THIS CHAPTER.

EO 001.01 CONFIGURE MHDS FOR SUBMARINE BROADCAST

1. Performance:

   Configure MHDS For Submarine Broadcast

2. Conditions:

   a. Given:

      (1) MHDS Terminal;

      (2) Keymat; and

      (3) References.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Trainer or alongside submarine

3. Standard: The trainee shall, IAW references, Configure MHDS For Submarine Broadcast by:

   a. Identifying broadcast port;

   b. Setting broadcast parameters;

   c. Activating broadcast port; and

   d. Testing broadcast port

4. Time.

   a. 1.6 x 50 minute periods

   b. EC x 1 x 50 minute period each trainee (practical)

   c. Total 7.6 x 50 min period based on maximum course load

5. Method / Media.

   a. Demonstration-practice

   b. Scenario based activity

6. References.

   C1, C2.

7. Training/Learning Aids.

   a. Screenshot; and

   b. PPT.

8. Test Details.

   a. This EO will be tested in EC 001.01 and PC 001.

9. Remarks.

Trainees will conduct a group scenario based activity on completion of teaching points of 40 min duration.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Identify broadcast port	Demo/Practice	10min	C1, C2
TP2	Set broadcast parameters	Demo/Practice	10min	C1, C2
TP3	Activate broadcast port	Demo/Practice	10min	C1, C2
TP4	Test broadcast port	Demo/Practice	10min	C1, C2
TP5	Setup broadcast	Scenario	40min	C1, C2

ENABLING CHECK – EO 001.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. MHDS Terminal.

Arrangements and Procedures

2. EC 001.01 will be a Practical assessment based on content from EO 001.01.
3. Pass/Fail
4. Time required: 1 x 50 min period
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 001.01.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. If you have any questions during this assessment, raise your hand.
4. References will be available.
5. Pass/Fail
6. You will have 1 x 50 min period to complete this assessment.
7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Identifying broadcast port			
Setting broadcast parameters			
Activating broadcast port			
Testing broadcast port			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature

Comments

The trainee must achieve SAT to pass this assessment.

EO 001.02 CONFIGURE MHDS FOR SUBMARINE SHIP/SHORE

1. Performance:

Configure MHDS For Submarine Ship/Shore

2. Conditions:

a. Given:

(1) MHDS Terminal; and

(2) References

b. Denied:

(1) Assistance

c. Environmental:

(1) Trainer or alongside Submarine

3. Standard: The trainee shall, IAW references, Configure MHDS for Submarine Ship/Shore by:

a. Identifying ship/shore port;

b. Setting ship/shore parameters;

c. Activating ship/shore port; and

d. Testing ship/shore port.

4. Time:

a. 1.6 x 50 minute periods

b. EC x 1 x 50 minute period each trainee (practical)

c. Total 7.6 x 50 minute periods based on maximum course load

5. Method / Media:

a. Demonstration-practice; and

b. Scenario based activity.

6. References:

C1, C2.

7. Training Aids:

a. Port screenshots; and

b. PPT.

8. Test Details.

a. This EO is tested in EC 001.02 and PC 001.

9. Remarks.

Trainees will conduct a group scenario based activity on completion of teaching points of 40 minutes duration.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Identify ship/shore port	Demo/Practice	10min	C1, C2
TP2	Set ship/shore parameters	Demo/Practice	10min	C1, C2
TP3	Activate ship/shore port	Demo/Practice	10min	C1, C2
TP4	Test ship/shore port	Demo/Practice	10min	C1, C2
TP5	Setup Ship/Shore circuit	Scenario	40min	C1, C2

ENABLING CHECK – EO 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 001.02 will be a practical test based on content from EO 001.02.
3. Pass/Fail
4. Time required: 1 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a Practical test based on content from EO 001.02.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. If you have any questions during the exam, raise your hand.
4. References will be available.
5. Pass/Fail
6. You will have 1 x 50 min period to complete this exam.
7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Identifying ship/shore port			
Setting ship/shore parameters			
Activating ship/shore port			
Testing ship/shore port			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

EO 001.03 CONFIGURE MHDS FOR SUBMARINE INTERSHIP

1. Performance: Configure MHDS for Submarine Intership

2. Conditions:

   a. Given:

      (1) MHDS Terminal; and

      (2) References.

   b. Denied:

      (1) Assistance

   c. Environmental: (1) Trainer; or

      (2) Submarine alongside.

3. Standard. The trainee shall, IAW references, Configure MHDS For Submarine Intership by:

   a. Identifying inter/ship port;

   b. Setting inter/ship parameters;

   c. Activating inter/ship port; and

   d. Testing inter/ship port

4. Time.

   a. 1.6 x 50 minute periods

   b. EC 1 x 50 minute period each trainee (practical)

   c. Total 7.6 x 50 min periods based on maximum course load

5. Method / Media.

   a. Demonstration-practice

   b. Scenario based activity

6. References.

   C1, C2.

7. Training Aids.

   a. Port screenshots; and

   b. PPT.

8. Test Details.

   This EO is tested in EC 001.03 and PC 001.

9. Remarks.

   Trainees will conduct a group scenario based activity on completion of teaching points of 40 min duration.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Identify intership port	Demo/Practice	10min	C1, C2
TP2	Set intership parameters	Demo/Practice	10min	C1, C2
TP3	Activate intership port	Demo/Practice	10min	C1, C2
TP4	Test intership port	Demo/Practice	10min	C1, C2
TP5	Setup intership circuit	Scenario	40min	C1, C2

ENABLING CHECK – EO 001.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside

Arrangements and Procedures

2. EC 001.03 will be a practical test based on content from EO 001.03.
3. Pass/Fail
4. Time required: 1 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a Practical test based on content from EO 001.03.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. If you have any questions during the exam, raise your hand.
4. References will be available.
5. Pass/Fail
6. You will have 1 x 50 min period to complete this exam.
7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST.

CHECK LIST	SAT	UNSAT	COMMENTS
Identifying inter/ship port			
Setting inter/ship parameters			
Activating inter/ship port			
Testing inter/ship port			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

EO 001.04 CONFIGURE MHDS FOR SUBMARINE SYSTEM BACKUP

1. Performance: Configure MHDS for Submarine System Backup

2. Conditions:

   a. Given:

      (1) MHDS terminal

      (2) Back-up software

      (3) Electronic media

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      MHDS trainer

3. Standard: The trainee shall, IAW references, Configure MHDS for Submarine System Backup by:

   a. Identifying media and backup software;

   b. Following software instructions;

   c. Storing backup medium; and

   d. Initialize backup software

4. Time.

   a. 3.3 x 50 min periods

   b. EC 2 x 50 min periods each trainee (practical)

   c. Total 15.3 x 50 min periods based on Maximum course load

5. Method / Media.

   a. Lecture;

   b. Demo;

   c. Practice; and

   d. Scenario.

6. References.

   C1, C2.

7. Training Aids.

   a. PPT;

   b. Backup software; and

   c. Electronic media.

8. Test Details.

a. Practical assessment

9. Remarks.

Trainees will conduct a group scenario based activity on completion of teaching points of 50 min duration.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Identify media and backup software	Lecture	25 min	C1,C2
TP2	Follow software instructions	Demo/Practice	30 min	C1,C2
TP3	Store backup medium	Demo/Practice	30 min	C1,C2
TP4	Test backup software	Demo/Practice	30 min	C1,C2
TP5	Initialize system backup	Scenario	50 min	C1,C2

ENABLING CHECK – EO 001.04

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 001.04 will be a Practical test based on content from EO 001.04.
3. Pass/Fail
4. Time required: 2 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.04 will be a Practical test based on content from EO 001.04.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. If you have any questions during the exam, raise your hand.
4. References will be available.
5. Pass/Fail
6. You will have 2 x 50 min period to complete this exam.
7. Do you have any questions before you start? PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Identify media and backup software			
Storing backup medium			
Testing backup software			
Following software instructions			
Initialize backup software			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. PC 001 will be a Practical test based on content from PO 001.
3. Pass/Fail
4. Time required: 4 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a Practical test based on content from PO 001
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. If you have any questions during the exam, raise your hand.
4. References will be available.
5. Pass/Fail
6. You will have 4 x 50 minute periods to complete this exam.
7. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Identifying broadcast port			
Setting broadcast parameters			
Activating Broadcast port			
Testing Broadcast port			
Identifying ship/shore port			
Setting ship/shore parameters			
Activating ship/shore port			
Testing ship/shore port			
Identifying inter/ship port			
Setting inter/ship parameters			
Activating inter/ship port			
Testing inter/ship port			
Identifying media and backup software			
Following software instructions			
identifying backup medium			
Initialize backup software			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

EO 002.01 IDENTIFY EQUIPMENT CAPABILITIES AND LIMITATIONS

1. Performance: Identify Equipment Capabilities and Limitations

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      Trainer

3. Standard. The trainee shall, IAW references, Identify Equipment Capabilities and Limitations by:

   a. Identifying VLF/LF frequency range and antenna;

   b. Identifying HF frequency range, power output and antenna;

   c. Identifying VHF frequency range, power output and antenna;

   d. Identifying UHF frequency range, power output and antenna; and

   e. Identifying Masts/antenna, Communication mast HF, Communication mast UHF, Periscopes, Emergency, Fin Loops, Types of communications, Frequency range, BWA.

4. Time.

   a. 2.5 x 50 min periods

   b. EC 1 x 50 min period (written)

   c. Total 3.5 x 50 min period base

5. Method / Media.

   a. Lecture

6. References.

   B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49

7. Training Aids.

   a. PPT

8. Test Details.

   a. This EO is tested in EC 002.01 and PC 002

9. Remarks.

   NIL

10. Teaching Points.

	Description	Method	Time	Reference
TP1	VLF/LF frequency range and antenna	Lecture	25 min	B18 – B49
TP2	HF frequency range, power output and antenna	Lecture	25 min	B18 – B49
TP3	VHF frequency range, power output and antenna	Lecture	25 min	B18 – B49
TP4	UHF frequency range, power output and antenna	Lecture	25 min	B18 – B49
TP5	Masts/AE, Communication mast HF, Communication mast UHF, Periscopes, Emergency, Fin Loops, Types of communications, Frequency range, BWA	Lecture	25 min	B18 – B49

ENABLING CHECK – EO 002.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.01 will be a written test based on content from EO 002.01.
3. Pass mark will be 80%.
4. Time required: 1 x 50 min period.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a written test based on content from EO 002.01.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during the exam, raise your hand.
8. References will not be provided.
9. Pass mark is 80%
10. You will have 1 x 50 min period to complete this exam.
11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Identify VLF/LF frequency range and antenna		
Identify HF frequency range, power output and antenna		
Identify VHF frequency range, power output and antenna		
Identify UHF frequency range, power output and antenna		
Identify Masts/antenna, Communication mast HF, Communication mast UHF, Periscopes, Emergency, Fin Loops, Types of communications, Frequency range, BWA		
Student Name		Student Signature
Assessor Name		Assessor Signature
Comments		

Trainee must achieve 80% to pass this assessment

EO 002.02 CONDUCT SUBMARINE BROADCAST PROCEDURES

1. Performance: Conduct Submarine Broadcast Procedures

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Communication trainer; or

      (2) Submarine alongside.

3. Standard. The trainee shall, IAW references, Conduct Submarine Broadcast Procedures by:

   a. Configuring HF broadcast to include:

      (1) Selecting antenna

      (2) Tuning receiver

      (3) Configuring modem

      (4) Loading crypto

      (5) Patching broadcast

      (6) Configuring broadcast port on MHDS

      (7) Identifying red and black patch panel

      (8) Explaining signal flow

   b. Configuring LF broadcast to include:

      (1) Selecting antenna

      (2) Tuning receiver

      (3) Configuring modem

      (4) Loading crypto

      (5) Patching broadcast

      (6) Configuring broadcast port on MHDS

      (7) Identifying red and black patch panel

      (8) Explaining signal flow

   c. Configuring SATCOM broadcast to include:

      (1) Selecting antenna

      (2) Tuning receiver

(3) Configuring modem

(4) Loading crypto

(5) Patching broadcast

(6) Configuring broadcast port on MHDS

(7) Identifying red and black patch panel

(8) Explaining signal flow

d. Describing Submarine PM/OTO's to include:

(1) Describing specific procedural messages and OTO's

(2) Describing call tape formats

4. Time.

a. 9.7 x 50 min periods

b. EC 2 x 50 min periods each trainee (practical)

c. Total 25.7 x 50 min periods based on maximum course load

5. Method / Media.

a. lecture;

b. Demo/practice; and

c. Scenario.

6. References.

A3, B3, B5, B6, B9, B10, B52, C1, C3, C4.

7. Training Aids.

8. Test Details.

a. This EO is tested in EC 002.02 and PC 002

9. Remarks.

a. NIL

10. Teaching Points.

	Description	Method	Time	Reference

TP1	Configure HF broadcast 1. Select antenna 2. Tune receiver 3. Configure modem 4. Load crypto 5. Patch broadcast 6. Configure broadcast port on CGS 7. Identify red and black patch panel 8. Explain signal flow	Demonstration/ Practice	15 min 15 min 20 min 20 min 15 min 20 min 15 min 25 min	A3, B3, B9, B10, B52, C1, C3, C4.
TP2	Configure LF broadcast 1. Select antenna 2. Tune receiver 3. Configure modem 4. Load crypto 5. Patch broadcast 6. Configure broadcast port on CGS 7. Identify red and black patch panel 8. Explain signal flow	Demonstration/ Practice	15 min 15 min 20 min 20 min 15 min 20 min 15 min 25 min	A3, B3, B9, B10, B52, C1, C3, C4.
TP3	Configure SATCOM broadcast 1. Select antenna 2. Tune receiver 3. Configure modem 4. Load crypto 5. Patch broadcast 6. Configure broadcast port on CGS 7. Identify red and black patch panel 8. Explain signal flow	Demonstration/ Practice	15 min 15 min 20 min 20 min 15 min 20 min 15 min 25 min	A3, B3, B9, B10, B52, C1, C3, C4.

TP4	Describe submarine procedural messages and OTO'S 1. Describe specific procedural messages and OTO's 2. Describe call tape formats	Demonstration/ Practice	25min 25 min	A3, B5, B6.

ENABLING CHECK – EO 002.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Communication Trainer or Submarine Alongside

Arrangements and Procedures

2. EC 002.02 will be a Practical test based on content from EO 002.02.
3. Pass/Fail
4. Time required: 2 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.02 will be a Practical test based on content from EO 002.02.
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass/Fail
5. You will have 2 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Configuring HF broadcast			
Configuring LF broadcast			
Configuring SATCOM broadcast			
Describing Submarine PM/OTO's			
Student Name	Pass	Fail	Student Signature
Student Name			Student Signature
Comments			

The trainee must achieve all SAT to pass this assessment.

EO 002.03 CONDUCT SHIP/SHORE PROCEDURES

1. Performance: Conduct Ship/Shore Procedures

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Communication trainer; or

      (2) Submarine alongside.

3. Standard. The trainee shall, IAW references, Conduct ship/shore procedures to include:

   a. Tuning HF transmit for submarine ship/shore by:

      (1) Powering on equipment

      (2) Loading crypto

      (3) Selecting antenna

      (4) Tuning transceiver

      (5) Configuring modem

      (6) Patching circuit

      (7) Configuring ship/shore port

      (8) Explaining signal flow

   b. Configuring DAMA for submarine ship/shore by:

      (1) Powering on equipment

      (2) Loading crypto

      (3) Selecting antenna

      (4) Tuning transceiver

      (5) Configuring modem

      (6) Patching circuit

      (7) Configuring ship/shore port

      (8) Explaining signal flow

c. Configuring RATT ship/shore answering by:

(1) Powering on equipment

(2) Selecting antenna

(3) Selecting ZKR frequencies

(4) Tuning receiver to frequency

(5) Configuring modem

(6) Patching to crypto device

(7) Configuring crypto device

(8) Patching to MHDS

4. Time.

a. 5.1 x 50 min periods

b. EC 4 x 50 min periods each trainee (practical)

c. Total 37.1 x 50 min periods based on maximum course load

5. Method / Media.

a. Practical;

b. Lecture; or

c. demonstration

6. References.

B19, B20, B30, B31, B33, B34, B35, B40, B45, B46, B52, C1, C3, C4.

7. Training Aids.

a. PPT

8. Test Details.

a. Practical Assessment

9. Remarks.

NIL

10. Teaching Points.

	Description	Method	Time	Reference

TP1	Tune HF transmit for submarine ship/shore 1. Power on equipment 2. Load crypto 3. Select antenna 4. Tune transceiver 5. Configure modem 6. Patch circuit 7. Configure ship/shore port 8. Explain signal flow	Lecture Demonstration Practice	5 min 20 min 5 min 10 min 20 min 5 min 10 min 10 min	B19, B20, B30, B31, B33, B34, B35, B40, B45, B46, B52.

TP2	Configure DAMA for submarine ship/shore 1. Power on equipment 2. Load crypto 3. Select antenna 4. Tune transceiver 5. Configure modem 6. Patch circuit 7. Configure ship/shore port 8. Explain signal flow	Lecture Demonstration Practice	5 min 20 min 5 min 10 min 20 min 5 min 10 min 10 min	B19, B20, B30, B31, B33, B34, B35, B40, B45, B46, B52, C1, C3, C4.

TP3	Configure RATT ship/shore answering 1. Power on equipment 2. Select antenna 3. Select ZKR frequencies 4. Tune receiver to frequency 5. Configure modem 6. Patch to crypto device 7. Configure crypto device 8. Patch to MHDS	Lecture Demonstration Practice	5 min 20 min 5 min 10 min 20 min 5 min 10 min 10 min	B19, B20, B30, B31, B33, B34, B35, B40, B45, B46, B52, C1, C3, C4.

ENABLING CHECK – EO 002.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Communications trainer or alongside submarine

Arrangements and Procedures

2. EC 002.03 will be a practical test based on content from EO 002.03.
3. Pass/Fail
4. Time required: 4 x 50 min period per trainee.
5. References will be available

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.03 will be a practical test based on content from EO 002.03.
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass/Fail
5. You will have 4 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Tuning HF transmit for submarine ship/shore			
Configuring DAMA for submarine ship/shore			
Configuring RATT ship/shore answering			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature

Comments

The trainee must achieve SAT to pass this assessment.

EO 002.04 CONDUCT SHIP/SHIP – SHIP/AIR PROCEDURES

1. Performance: Conduct Ship/Ship and Ship/Air Procedures

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Communication trainer; or

      (2) Submarine alongside

3. Standard. The trainee shall, IAW references, Conduct ship/ship and ship/air procedure by:

   a. Configuring HF for secure voice to include:

      (1) Powering on equipment

      (2) Selecting antenna

      (3) Tuning transceiver

      (4) Patching to crypto device

      (5) Loading crypto

      (6) Remoting to user position

   b. Configuring for HF CRATT to include:

      (1) Powering on equipment

      (2) Selecting antenna

      (3) Tuning transceiver

      (4) Configuring modem

      (5) Patching to crypto device

      (6) Loading crypto device

   c. Configuring UHF LOS CRATT to include:

      (1) Powering on equipment

      (2) Selecting antenna

      (3) Configuring transceiver/receiver

      (4) Tuning transceiver/receiver

      (5) Configuring modem

      (6) Configuring crypto

d. Configuring HF for plain voice to include:

(1) Powering on equipment

(2) Selecting antenna

(3) Tuning HFTR

(4) Remoting to user position

e. Configuring UHF for LOS secure voice to include:

(1) Powering on equipment

(2) Selecting antenna

(3) Configuring transceiver/receiver

(4) Tuning transceiver

(5) Loading crypto

(6) Patching to crypto

f. Configuring UHF for LOS plain language voice to include:

(1) Powering on equipment

(2) Selecting antenna

(3) Configuring transceiver/receiver

(4) Tuning transceiver

(5) Remoting to user position

g. Configuring ECB's to include:

(1) Mounting ECB into program

(2) Configuring ECB program

(3) Patching ECB

(4) Inputting signal into ECB

h. Configuring DAMA for satellite secure voice to include:

(1) Powering on equipment

(2) Loading Crypto

(3) Selecting antenna

(4) Configuring transceiver/receiver

(5) Tuning transceiver/receiver

(6) Patching circuit

i. Configuring VHF to include:

(1) Powering on equipment

(2) Selecting antenna

(3) Selecting channel (4) Remoting to user position

4. Time.

   a. 8.6 x 50 min periods

   b. EC 8 x 50 min periods each trainee (practical)

   c. Total 56.6 x 50 min periods based on maximum course load

5. Method / Media.

   a. Practical;

   b. Scenario; or

   c. Practical demonstration onboard submarine

6. References.

   A3, B4, B7, B9, B16, B17, B19, B20, B21, B22, B28, B31, B33, B34, B40, B41, B45, B46, B47, B52, C1, C3, C4, C5.

7. Training Aids.

   a. PPT

8. Test Details.

   a. Practical Assessment

9. Remarks.

   a. One day practical scenario based in trainer or submarine alongside.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Configure HF for secure voice 1. Power on equipment 2. Select antenna 3. Tune transceiver 4. Patch crypto device 5. Load crypto 6. Remote to user position	Lecture Demonstration Practice	10 Min 5 min 20 min 5 min 20 min 10 min	A3, B9, C1, C3, C4, B19, B20, B21, B22, B31, B33, B34, B40, B40.

TP2	Configure for HF CRATT 1. Power on equipment 2. Select antenna 3. Tune transceiver 4. Configure modem 5. Patch crypto device 6. Load crypto	Lecture Demonstration Practice	10 min 5 min 10 min 15 min 5 min 20 min	B19, B20, B30, B31, B33, B34, B35, B40, B45, B46, B52.

TP3	Configure UHF LOS CRATT 1. Power on equipment 2. Select antenna 3. Configure transceiver/receiver 4. Tune transceiver/receiver 5. Configure modem 6. patch crypto device 7. load crypto device	Lecture Demonstration Practice	5 min 5 min 10 min 10 min 15 min 5 min 15 min	A3, B4, B19, B20, B30, B31, B33, B34, B35, B40, B43.
TP4	Configure HF for plain voice 1. Power on equipment 2. Select antenna 3. Tune HFTR 4. Remote to user position	Lecture Demonstration Practice	5 min 5 min 10 min 10 min	A3, A4, B19, B20, B30, B31, B33, B34, B35, B40, B43.

TP5	Configure UHF for LOS secure voice 1. Power on equipment 2. Select antenna 3. Configure transceiver/receiver 4. Tune transceiver 5. Load crypto 6. Patch to crypto	Lecture Demonstration Practice	5 min 5 min 10 min 10 min 15 min 5 min	A3, B4, B19, B20, B31, B33, B34, B35, B40, B43, C3, C4.
TP6	Configure UHF for LOS plain language voice 1. Power on equipment 2. Select antenna 3. Configure transceiver/receiver 4. Tune transceiver 5. Patch to user position	Lecture Demonstration Practice	5 min 5 min 10 min 10 min 5 min	A3, B4, B19, B20, B31, B33, B34, B35, B40, B43, C3, C4.
TP7	Configure ECB'S 1. Mount ECB into program 2. Configure ECB program 3. Patch ECB 4. Input signal into ECB	Lecture Demonstration Practice	10 min 10 min 10 min 20 min	A3, B7, B9, B16, B41, B47.

TP8	Configure DAMA for satellite secure voice 1. Power on equipment 2. Load Crypto 3. Select antenna 4. Configure transceiver/receiver 5. Tune transceiver/receiver 6. Patch circuit	Lecture Demonstration Practice	5 min 15 min 5 min 10 min 10 min 5 min	A3, B4, B19, B20, B31, B33, B34, B35, B40, B43, C3, C4.
TP9	Configure VHF 1. Power on equipment 2. Select antenna 3. Select channel 4. Remote to user position	Lecture Demonstration Practice	5 MIN 5 MIN 10 MIN 5 MIN	B4, B7, B16, B17, B22, B28, B34, B43, B44, B48, B49, C5.

ENABLING CHECK – EO 002.04

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Communications trainer or alongside submarine

Arrangements and Procedures

2. EC 002.04 will be a practical test based on content from EO 002.04.
3. Pass/Fail
4. Time required: 8 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.04 will be a practical test based on content from EO 002.04.
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass/Fail
5. You will have 4 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Configuring HF for secure voice			
Configuring for HF CRATT			
Configuring UHF LOS CRATT			
Configuring HF for plain voice			
Configuring UHF for LOS secure voice			
Configuring UHF for LOS plain language voice			
Configuring DAMA for satellite secure voice			
Configuration of the ECB			

Configuring VHF			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

6. Trainer or submarine alongside.

Arrangements and Procedures

7. PC 002 will be a Practical test based on content from PO 001.
8. Pass/Fail
9. Time required: 4 x 50 min period per trainee.
10. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

8. PC 002 will be a Practical test based on content from PO 002
9. If you have any questions during the exam, raise your hand.
10. References will be available.
11. Pass/Fail
12. You will have 4 x 50 minute periods to complete this exam.
13. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Configuring HF for secure voice			
Configuring for HF CRATT			
Configuring UHF LOS CRATT			
Configuring HF for plain voice			
Configuring UHF for LOS secure voice			

Configuring UHF for LOS plain language voice			
Configuring DAMA for satellite secure voice			
Configuration of the ECB			
Configuring VHF			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

EO 003.01 DISSEMINATE KEY INFORMATION

1. Performance: Disseminate Key Information

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance

   c. Environmental:

      (1) Trainer; and

      (2) Submarine at sea or alongside.

3. Standard. The trainee shall, IAW references, disseminate key information by:

   a. Identifying submarine specific content.

   b. Utilizing references to decode fields

4. Time.

   a. 2 x 50 min periods

   b. EC 1 x 50 min period

   c. Total 3 x 50 min periods

5. Method / Media.

   a. Lecture; and

   b. Scenario

6. References.

   A3, B9, B10, B11, B12, B13, B15.

7. Training Aids.

   a. Training messages; and

   b. PPT.

8. Test Details.

   a. Based on the content or EO 003.01 and PO 001

9. Remarks.

   a. NIL

Teaching Points.

	Description	Method	Time	Reference

TP1	Identify submarine specific content	Lecture	50 min	A3, B9, B10, B11, B12, B13, B15.
TP2	Utilize references to decode fields	Lecture	50 min	A3, B9, B10, B11, B12, B13, B15.

ENABLING CHECK – EO 003.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 003.01 will be a written test based on content from EO 003.01.

3. Pass mark will be 80% 4. Time required: 1 x 50 min period.

5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a written test based on content from EO 003.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be available.

9. Pass mark is 80%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Identifying submarine specific content		
Utilize references to decode fields		
Student Name		Student Signature

Assessor Name and signature		Assessor Signature
Comments		

Trainee must achieve 80% to pass this assessment

EO 003.02 DRAFT SUBMARINE FORMATTED MESSAGES

1. Performance: Drafting Submarine Formatted Messages

2. Conditions:

   a. Given:

      (1) References

      (2) Messages

      (3) Message drafting software; and

      (4) State boards

   b. Denied:

      (1) Supervision; and

      (2) Assistance

   c. Environmental: (1) Trainer; or

      (2) Submarine alongside.

3. Standard. The trainee shall, IAW references, draft submarine formatted messages by:

   a. Identifying submarine specific messages; and

   b. Obtaining message content.

4. Time.

   a. 2 x 50 min periods

   b. EC 1 x 50 min period

   c. Total 3 x 50 min periods

5. Method / Media.

   a. Lecture; and

   b. Scenario.

6. References.

   A3, B9, B10, B11, B12, B13, B15.

7. Training Aids.

   a. PPT

8. Test Details.

   a. Based on content of EO 003.02 and PC 003

9. Remarks.

   a. 50 min review scenario upon completion.

10. Teaching Points.

	Description	Method	Time	Reference

TP1	Submarine specific messages	Lecture and Scenario	50 min	A3, B9, B10, B11, B12, B13, B15.
TP2	Obtain message content	Lecture and Scenario	50 min	A3, B9, B10, B11, B12, B13, B15.

ENABLING CHECK – EO 003.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 003.02 will be a written test based on content from EO 003.02.

3. Pass mark will be 80% 4. Time required: 1 x 50 min period

5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.02 will be a written test based on content from EO 003.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be available.

9. Pass mark is 80%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Identifying submarine specific messages		
Obtaining message content		
Student Name		Student Signature

Assessor Name		Assessor Signature
Comments		

Trainee must achieve 80% to pass this assessment

PERFORMANCE CHECK – PO 003

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 003 will be a written test based on content from PO 003
3. Pass mark will be 80% 4. Time required: 1 x 50 min period
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a Written test based on content from PO 003
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during the exam, raise your hand.
8. References will be available.
9. Pass mark is 80%
10. You will have 1 x 50 min period to complete this exam.
11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Identifying submarine specific messages		
Obtaining message content		
Student Name		Student Signature
Assessor Name		Assessor Signature

Comments

The trainee must achieve all 80% to pass this assessment.

EO 004.01 FORMULATE RECOMMENDATIONS TO COMMAND

1. Performance: Formulate Recommendations to Command.

2. Conditions:

   a. Given:

      (1) State boards;

      (2) Submarine radio equipment; and

      (3) Pre-sail checklist.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Trainer; or

      (2) Submarine alongside

3. Standard. The trainee shall, IAW references, formulate recommendations to Command by:

   a. Analysing equipment to include:

      (1) Recognizing deficiencies

      (2) Isolating u/s equipment

      (3) Liaising with W Eng Tech Communications

      (4) Trouble shooting equipment

   b. Obtaining command priorities to include:

      (1) Identifying options (2) Analyzing options

4. Time.

   a. 2 x 50 min periods

   b. EC 2 x 50 min periods

   c. Total 4 x 50 min periods

5. Method / Media.

   a. Guided discussion; and

   b. Lecture

6. References.

   A3, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, C1, C2, C3, C4.

7. Training Aids.

   a. PPT; and

b. Scenario.

8. Test Details.

a. Based on content of EO 004.01 and PO 004

9. Remarks.

a. Review guided discussion 50 min.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Analyse equipment (1) Recognize deficiencies (2) Isolate u/s equipment (3) Liaise N/W Eng Tech Communications (4) Trouble shoot equipment	Lecture Guided discussion	50 min	A3, B18 – B49, C1, C2, C3, C4.
TP2	Obtain command priorities (1) Identify options (2) Analyze options	Lecture Guided discussion	50 min	A3, B18 – B49, C1, C2, C3, C4.

ENABLING CHECK – EO 004.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 004.01 will be a practical test based on content from EO 004.01.
3. Pass/Fail
4. Time required: 2 x 50 min period per trainee.
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.01 will be a practical test based on content from EO 004.01.
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass/Fail
5. You will have 2 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Analysing equipment			
Obtaining command priorities			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

EO 004.02 DELIVER REPORTS

1. Performance: Deliver Reports

2. Conditions:

   a. Given:

      (1) State boards;

      (2) References;

      (3) Logs; and

      (4) Submarine radio equipment.

   b. Denied:

      (1) Supervision; and

      (2) Assistance

   c. Environmental:

      (1) Trainer; or

      (2) Submarine alongside

3. Standard. The trainee shall, IAW references, deliver reports by:

   a. Delivering oral reports to include:

      (1) Gathering required information for submarine specific reports

   b. Delivering written reports to include:

      (1) Logging Ship/shore

      (2) Logging broadcast

4. Time.

   a. 2 x 50 min periods

   b. EC 1 x 50 min period

   c. Total 3 x 50 min periods

5. Method / Media.

   a. Lecture; and

   b. Question & answer

6. References.

   A3, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, C1, C2, C3, C4.

7. Training Aids.

   a. PPT

8. Test Details.

   a. Based on content of EO 004.02 and PO 004 Practical assessment

9. Remarks.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Deliver oral reports - Gather required info for submarine specific report	Interactive Lecture including review	50 min	A3, B18 – B49, C1, C2, C3, C4.
TP2	Deliver written reports • Log ship/shore • Log broadcast	Interactive Lecture including review	50 min	A3, B18 – B49, C1, C2, C3, C4.

ENABLING CHECK – EO 004.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 004.02 will be a written test based on content from EO 004.02.
3. Pass is 80%.
4. Time required: 1 x 50 min period
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.02 will be a written test based on content from EO 004.02.
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass is 80%.
5. You will have 1 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Deliver oral reports		
Deliver written reports		
Student Name		Student Signature
Assessor Name		Assessor Signature
Comments		

The trainee must achieve all 80% to pass this assessment.

PERFORMANCE CHECK – PO 004

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. PC 004 will be a scenario based written test based on content from PO 004
3. Pass is 80%.
4. Time required: 1 x 50 min period
5. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 will be a scenario based written test based on content from PO 004
2. If you have any questions during the exam, raise your hand.
3. References will be available.
4. Pass is 80%.
5. You will have 1 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	% WEIGHT	COMMENTS
Analysing equipment		
Obtaining command priorities		
“Recognizing Deficiencies” not tested		
Delivering oral reports		
Delivering written reports		
Student Name		Student Signature
Assessor Name		Assessor Signature
Comments		

The trainee must achieve 80% to pass this assessment.

EO 005.01 UTILIZE ATMOSPHERIC MONITORING EQUIPMENT

1. Performance: Utilize Atmospheric Monitoring Equipment

2. Conditions:

   a. Given:

      (1) References

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) Communications trainer; or

      (2) Submarine alongside or at sea

3. Standard. The trainee shall, IAW references, utilize atmospheric monitoring equipment by:

   a. Determining time for atmospheric monitoring; and

   b. Employing atmospheric monitoring equipment.

4. Time.

   a. 2 x 50 min periods

   b. EC 1 x 50 min period

   c. Total 3 x 50 min period

5. Method / Media.

   a. Lecture; and

   b. Practical

6. References.

   B53, B54.

7. Training Aids.

   a. PPT; and

   b. Atmospheric monitoring equipment

8. Test Details.

   a. EC will be oral confirmation questions; and

   b. Based on content of 005.01 and PC 005

9. Remarks.

10. Teaching Points.

	Description	Method	Time	Reference

TP1	Determine time for atmospheric monitoring	Lecture	50 min	B53, B54.
TP2	Employ atmospheric monitoring equipment	Practical	50 min	B53, B54.

ENABLING CHECK – EO 005.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 005.01 will be oral confirmation based on content from EO 005.01.
3. Pass/Fail
4. Time required: 1 x 50 min period
5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.01 will be an oral confirmation based on content from EO 005.01.
2. Pass/fail

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Determining time for atmospheric monitoring			
Employ atmospheric monitoring equipment			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve all SAT to pass this assessment.

EO 005.02 DISSEMINATE ATMOSPHERIC RESULTS

1. Performance: Disseminate Atmospheric Results

2. Conditions:

a. Given:

(1) Atmospheric testing equipment

b. Denied:

(1) Supervision; and

(2) Assistance.

c. Environmental:

(1) Classroom

(2) Trainer; and

(3) Submarine at sea or alongside.

3. Standard. The trainee shall, IAW references, disseminate atmospheric results by:

a. Recording atmospheric monitoring results to include: (1) Updating state board; and

(2) Updating logs.

4. Time.

a. 1 x 50 min period

b. EC 1 x 50 min period

c. Total 2 x 50 min period

5. Method / Media.

a. Lecture

6. References.

B53, B54.

7. Training Aids.

a. PPT

8. Test Details.

a. Oral Confirmation

9. Remarks.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Record atmospheric monitoring results	Lecture	50 min	B53, B54.

ENABLING CHECK – EO 005.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 005.02 will be an oral confirmation based on content from EO 005.02.
3. Pass/Fail
4. Time required: 50 min period
5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.02 will be an oral confirmation based on content from EO 005.02.

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Record atmospheric monitoring results			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

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Trainee must achieve Sat to pass this assessment

EO 005.03 ADVISE COMMAND ON ATMOSPHERIC RESULTS

1. Performance: Advise Command on Atmospheric Results

2. Conditions:

   a. Given:

      Atmospheric results

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      Trainer or submarine alongside.

3. Standard. The trainee shall, IAW references, advise command on atmospheric results by reporting verbally on atmospheric results

4. Time.

   a. 1 x 50 min period

   b. 1 x 50 min period

   c. Total 2 x 50 min period

5. Method / Media.

   a. Case study

6. References.

   B53, B54.

7. Training Aids.

   a. PPT

8. Test Details.

   a. Oral Confirmation questions O/C

9. Remarks.

   a. Nil

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Provide verbal reports on atmospheric results	Case Study	50 min	B53, B54.

ENABLING CHECK – EO 005.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer or submarine alongside.

Arrangements and Procedures

2. EC 005.03 will be an oral confirmation based on content from EO 005.03.
3. Pass/Fail
4. Time required: 1 x 50 min period
5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.03 will be an Oral Confirmation based on content from EO 005.03.

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Conduct verbal reports on atmospheric results			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

Trainee must achieve SAT to pass this assessment

PERFORMANCE CHECK – PO 005

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

6. Trainer or submarine alongside.

Arrangements and Procedures

7. PC 005 will be a practical exam based on content from PO 005
8. Pass/Fail.
9. Time required: 1 x 50 min period
10. References will be available.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

7. PC 005 will be a oral confirmation based on content from PO 005
8. If you have any questions during the exam, raise your hand.
9. References will be available.
10. Pass/Fail.
11. You will have 25 min period to complete this exam.
12. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Conduct verbal reports on atmospheric results			
Student Name	Pass	Fail	Student Signature
Assessor Name			Assessor Signature
Comments			

The trainee must achieve SAT to pass this assessment.

ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C =Commercial Publication

A1.	VCSSO's
A2.	CFCD 106 - Maritime Tactical Instructions
A3.	CFCD 107 - Maritime Command Communications Orders
A4.	CFNOS Communications School Satellite Communcations Info Package
A5.	CIS/01/203
A6.	JTFA CONPLAN SUBSMASH
A7.	CFCD 103 – Submarine Escape and Rescue Instructions
B1.	ACP 117 - Allied Routing Indicator Book
B2.	ACP 117 - CANSUPP
B3.	ACP 121 - Communcations General
B4.	ACP 125 - Radiotelephone Procedures
B5.	ACP 127 - Communications Instructions – Tape Relay Procedures
B6.	ACP 131 - Operating Signals
B7.	ACP 135 - Distress Communications
B8.	ACP 165 - Operational Brevity Codes
B9.	ACP 176 - Allied Naval and Maritime Air Communcations Instructions

B10.	ACP 176 - NATO SUPP 1 - NATO and Maritime Communications Instructions/Organization
B11.	APP 1 (NAVY/AIR) - Allied Maritime Voice Reporting Procedures
B12.	APP 4 VOL I - Allied Maritime Tactical Messages
B13.	APP 4 VOL II - Allied Maritime Tactical Messages

B14.	APP 7 - NATO Brevity Codes
B15.	ATP 1 VOL I - Tactical Instructions
B16.	ATP 10 - NATO Search and Rescue
B17.	ATP 57 - NATO SUBSAR
B18.	BRF 1966(27) - BWA Index
B19.	BRF 1966(27)01 - Aerial Transfer System
B20.	BRF 1966(27)01 - Aerial Transfer Equipment
B21.	BRF 1966(31) - Internal Communications Index
B22.	BRF 1966(31)01 - Internal Communications Equipment
B23.	BRF 2475 - Sub VLF/LF Aerial Outfit Fin and Buoy Mounted
B24.	BRF 2475(7) - Sub VLF/LF Aerial Outfit Fin and Buoy Mounted
B25.	BRF 2475(58) - Sub VLF/LF Aerial Outfit Fin and Buoy Mounted
B26.	BRF 4183 - VLF/LF Buoyant Cable Aerial Outfit
B27.	BRF 4183(8) - VLF/LF Buoyant Cable Aerial Outfit - Modifications
B28.	BRF 4567(113) - Voice Internal Communications System for S/M's
B29.	BRF 8112(8) - Transceiver Type H4640 - Modifications
B30.	BRF 8113(8) - 250 Watt HF Amplifier Type H1071 - Modifications
B31.	BRF 8115(8) - Terminal Selection Unit
B32.	BRF 8116(8) - Remove Voice Control Unit - Mod 5
B33.	BRF 8117(8) - Aerial Exchange outfit EUC - Mods
B34.	BRF 8225(4A) - Outfit AJU(1)/(2) - Integrated Comms Mast
B35.	BRF 8293 - Outfit EUD
B36.	BRF 8385A(4A) - VLF/LF Multichannel Transmission/Reception System Outfits CFCD -
B37.	BRF 8385A(8) - VLF/LF Multichannel Transmission/Reception System Outfits CFCD - Modification

B38.	BRF 8385B(8) - VLF/LF Multichannel Transmission/Reception System – Receiver 723-3104 Receiver Modifications

B39.	BRF 8673(8) - NAVAID Antenna Switching Unit Modification
B40.	BR 222(SM) - Communications Equipment Handbook (Submarine)
B41.	BR 241 - Submarine Escape and Rescue Handbook
B42.	BR 4112 - Handbook for Submarine Indicator Buoy Type 639
B43.	BR 8225 - Integrated Communications Mast AJU
B44.	BR 8356 - Periscope Aerial Outfits AVS and AVT
B45.	BR 8385A - Outfit CFD
B46.	BR 8385B - Outfit CFD
B47.	BR 8518 - Expendable Communications Buoy System for Submarines
B48.	BR 8628 - Periscope Aerial Outfit AVS/AVT
B49.	BR 8653 - VHF Transceiver Type 1452
B50.	MATELO Handbook
B51.	NAMAR COMPLAN
B52.	RNCP 1 - Royal Navy Communications Organization
B53.	BR 1326 - Air Purification in Submarines
B54.	BR1326(A) - Air Purification in Submarines
C1.	CGS-100 Operator Manual
C2.	Miscellaneous Software User Guides
C3.	Mini DAMA Quick Reference Guide
C4.	Mini DAMA Handbook
C5.	ICOM VHF Handbook

ANNEX B – TASK LIST

JBS/SS TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
T0001	Conduct Submarine Ship/Shore Ship/Air Procedures	T	002
T0002	Conduct Submarine Broadcast Procedures	T	002
T0003	Operate Submarine Communications Equipment	T	002
T0004	Perform Submarine COMMCEN Functions	T	003
T0005	Operate Submarine Atmospheric Monitoring	T	005
NT0006	Configure Submarine MHDS	T	001
NT0007	Configure MHDS for Submarine System Backup	T	001
NT0008	Advise Command on Status of Submarine Communications	T	004

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

V2.1 Dated 3 Feb 2021 - A change in rank names was actioned IAW CANFORGEN 112/20 RCN Junior Rank Change Instructions.

Dates: 05 – 09 MARCH 2012 Location: CFNOS HALIFAX BLDG S17
Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Hiebert	MGR	Lt(N)	00207	HMCS CORNERBROOK
Robinson	RD	Lt	00211	CFNOS STDS
Smith	KP	Lt	00211	JAMAICA DEFENCE FORCE
Chiasson	M	PO1	00299	MARLANTHQ N32
MacPhee	B	PO1	00299	PCC ESQUIMALT
Patry	F	PO1	00366	CFNOS STDS
Chapman	W	PO2	00299	CANFLTLANT HQ
Gagnon	R	PO2	00299	MOG5 HQ
Pierangeli	TL	PO2	00299	DMTE 2-2-4
Lindstrom	J	S1	00299	HMCS CHICOUTIMI
Houde	SLP	S2	00299	HMCS ATHABASKAN

Board Objective: Rewrite QSP for Submarine Nav Comm
Specification Task – No Train Decisions:

The board decided that all tasks in the SS were required to be trained.

Discussions / Decisions:

1. Board convened at 0900 05 Mar 2012 with the Board Chairman introducing himself and the members in turn. The Board Chairman outlined briefly his responsibilities, with the TDO continuing with a more thorough brief on Board Members duties and responsibilities.

2. The board then commenced the analysis portion. During the Task Analysis the following recommendations were made:

Board recommended implementing a New SS Task PO 006 Perform Duties of ISSO for PO2's as they will be expected to perform these duties on board. The board recommends either a follow-on course for this section or to provide the training by placing members on that section of CISN Manager Course when available.

The board recommends that Off-Line Encryption remain uninstructed. VIC class submarines are fitted for but not with off-line encryption equipment and it will be a mission fit in the future for submarines which sail on NATO deployments. It is the Boards' recommendation that this be trained as required for missions.

3. The board reconvened on 06 Mar, where the Analysis portion was finalized and the Design phase commenced with Target Population Analysis. In the afternoon the board commenced IA Scalaring.

4. On 07 Mar, the board continued with IA Scalaring and commenced assigning references. In the afternoon it commenced the Learning Assessment Plan.

5. On 08 Mar, commenced writing lesson specifications and drafting teaching points.

6. On 09 Mar, reviewed document and adjourned.

Lessons Learned:

1. Familiarity with previous QSP documentation is essential prior to starting a board.

2. Board members should remain consistent throughout the duration.

3. The Board was challenged to complete the writing of the QSP document within one week.

Follow-up Action:

1. CFNOS Submarine Division is responsible for writing EC and PC checklists;

2. SPQR documentation for changes to this SS are to be completed by CFNOS Standards;

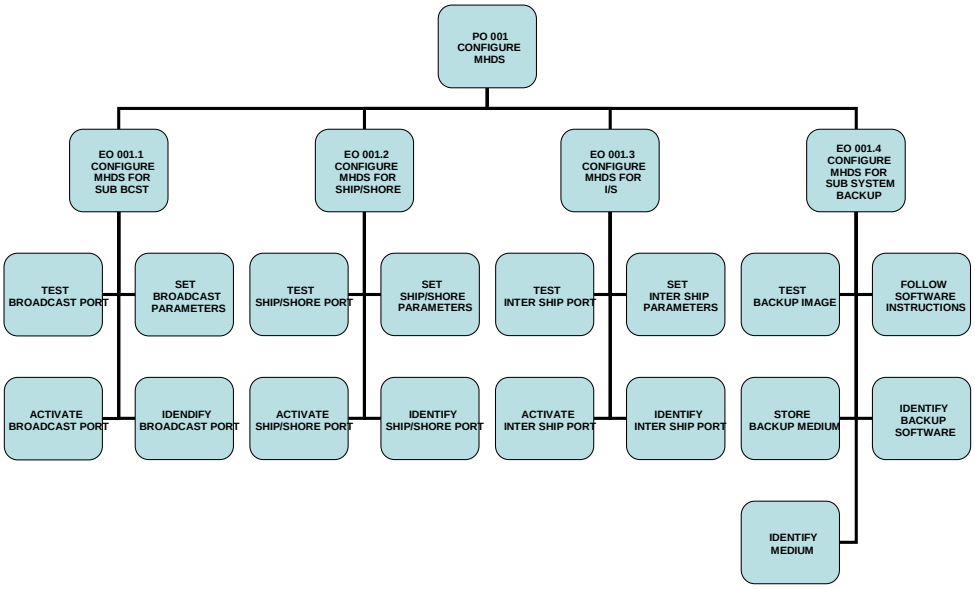
3. New SS to be created for "Perform Duties of ISSO Onboard a Submarine" to be completed by DMTE.

Other Considerations (if applicable)

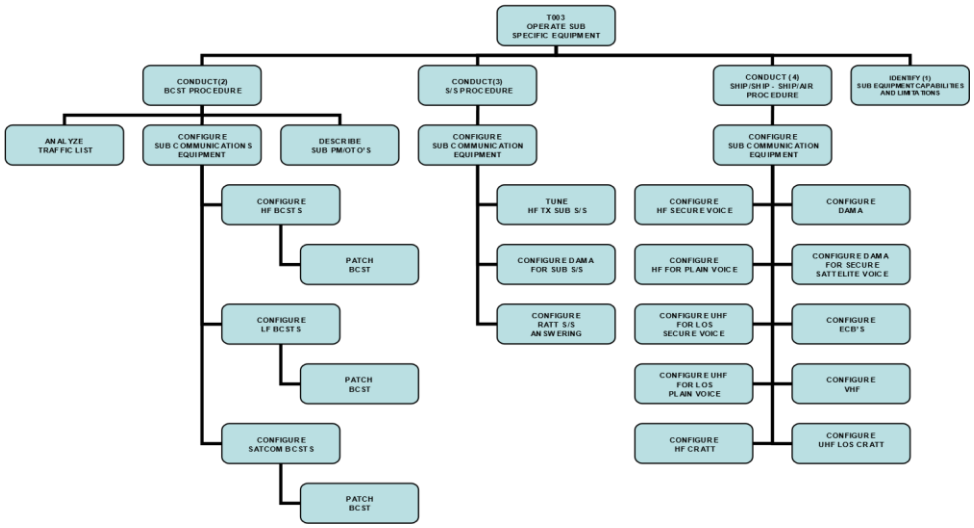
14 Mar 2012, AIIW SS document completed. It will be submitted with the SQPR (completed by TDO Lt Robinson, CFNOS STDS), updated AIIW QSP, along with a covering letter to DMTE.

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR

PO001



PO002


T003
OPERATE SUB
SPECIFIC EQUIPMENT
CONDUCT(2)
BCST PROCEDURE
CONDUCT(3)
S/S PROCEDURE
CONDUCT (4)
SHIP/SHIP - SHIP/AIR
PROCEDURE
IDENTIFY (1)
SUB EQUIPMENT CAPABILITIES
AND LIMITATIONS
ANALYZE
TRAFFIC LIST
CONFIGURE
SUB COMMUNICATIONS
EQUIPMENT
DESCRIBE
SUB PM/OTO'S
CONFIGURE
SUB COMMUNICATION
EQUIPMENT
CONFIGURE
SUB COMMUNICATION
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HF BCSTS
PATCH
BCST
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LF BCSTS
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SATCOM BCSTS
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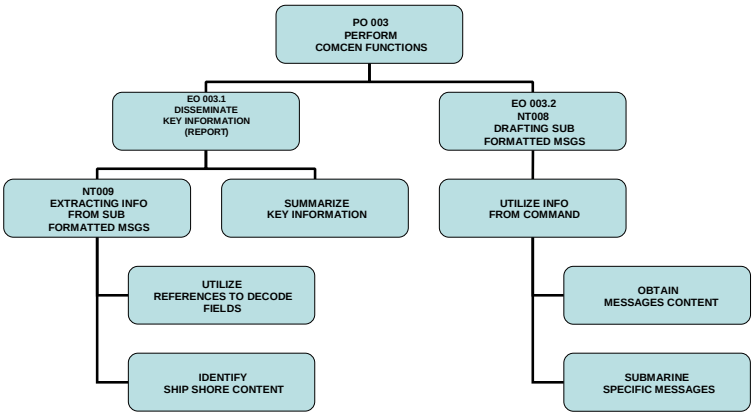
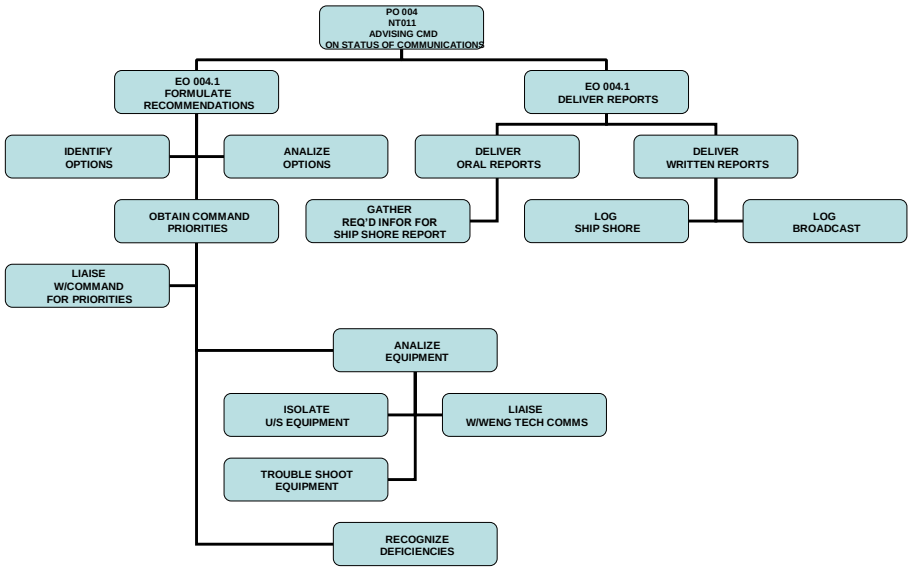
BCST
TUNE
HF TX SUB S/S
CONFIGURE DAMA
FOR SUB S/S
CONFIGURE
RATT S/S
ANSWERING
CONFIGURE
HF SECURE VOICE
CONFIGURE
HF FOR PLAIN VOICE
CONFIGURE UHF
FOR LOS
SECURE VOICE
CONFIGURE UHF
FOR LOS
PLAIN VOICE
CONFIGURE
HF CRATT
CONFIGURE
DAMA
CONFIGURE DAMA
FOR SECURE
SATTELITE VOICE
CONFIGURE
ECB'S
CONFIGURE
VHF
CONFIGURE
UHF LOS CRATT

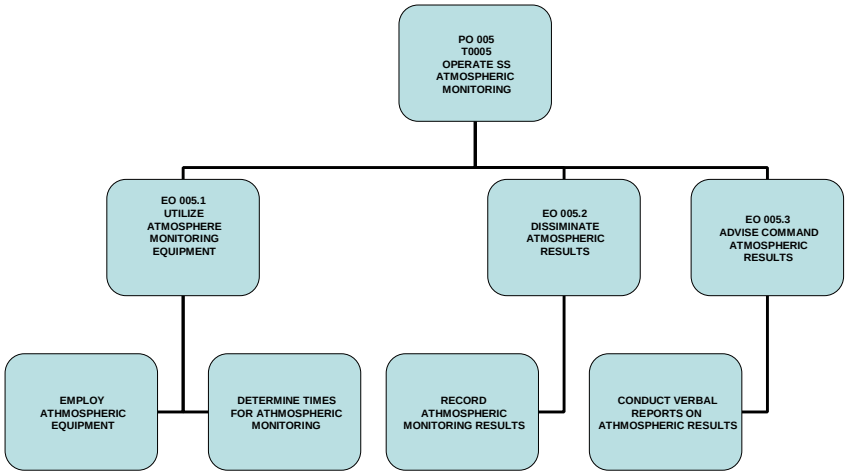
PO003


PO 003
PERFORM
COMCEN FUNCTIONS
EO 003.1
DISSEMINATE
KEY INFORMATION
(REPORT)
EO 003.2
NT008
DRAFTING SUB
FORMATTED MSGS
NT009
EXTRACTING INFO
FROM SUB
FORMATTED MSGS
SUMMARIZE
KEY INFORMATION
UTILIZE INFO
FROM COMMAND
UTILIZE
REFERENCES TO DECODE
FIELDS
IDENTIFY
SHIP SHORE CONTENT
OBTAIN
MESSAGES CONTENT
SUBMARINE
SPECIFIC MESSAGES

PO004


PO 004
NT011
ADVISING CMD
ON STATUS OF COMMUNICATIONS
EO 004.1
FORMULATE
RECOMMENDATIONS
EO 004.1
DELIVER REPORTS
IDENTIFY
OPTIONS
ANALIZE
OPTIONS
DELIVER
ORAL REPORTS
DELIVER
WRITTEN REPORTS
OBTAIN COMMAND
PRIORITIES
GATHER
REQ'D INFOR FOR
SHIP SHORE REPORT
LOG
SHIP SHORE
LOG
BROADCAST
LIAISE
W/COMMAND
FOR PRIORITIES
ANALIZE
EQUIPMENT
ISOLATE
U/S EQUIPMENT
LIAISE
W/WENG TECH COMMS
TROUBLE SHOOT
EQUIPMENT
RECOGNIZE
DEFICIENCIES

PO005


PO 005
T0005
OPERATE SS
ATMOSPHERIC
MONITORING
EO 005.1
UTILIZE
ATMOSPHERE
MONITORING
EQUIPMENT
EO 005.2
DISSIMINATE
ATMOSPHERIC
RESULTS
EO 005.3
ADVISE COMMAND
ATMOSPHERIC
RESULTS
EMPLOY
ATHMOSPHERIC
EQUIPMENT
DETERMINE TIMES
FOR ATHMOSPHERIC
MONITORING
RECORD
ATHMOSPHERIC
MONITORING RESULTS
CONDUCT VERBAL
REPORTS ON
ATHMOSPHERIC RESULTS

ANNEX E – ON-JOB PERFORMANCE RECORD

(CERTIFICATE OF QUALIFICATION)

NAME and INITIALS RANK SN

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RESUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. The Training PO shall insert an "E" (Exception) in the remarks column of the OJPR and include a brief explanation opposite that PO.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

5. Upon completion of the OJPR sheet (trainee has completed all required POs), the following action will be taken:

   a. Fleet Exam (if applicable):

   b. Nominate the trainee (by message) for fleet examination:

      (1) Forward a completed copy of the Annex C (section 7 completed) to CFNOS Standards Section (as appropriate) at least five (5) days prior to examination date; and

      (2) Upon successful completion of the Fleet Exam, the Standards Section is to forward the trainee results by message to the parent unit and DPCOR.

   c. Administrative (all candidates):

      (1) The Divisional Officer shall complete section 8 of the OJPR Certificate of Qualification;

      (2) The individual's Qualification Record Sheet (QRS) shall be completed as described in A-P9-000-015/PT-000 and distributed to the respective MARLANT/MARPAC Headquarters for action; and

      (3) The OJPR shall be retained on the individual's UER until update action is reflected on the Personal Record Resume.

PROGESS REVIEW

6. Progress review shall be carried out quarterly by the Divisional Officer and [Section Head (NCMs) OR Training Officer (Officers)] .

OJPR REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE

7. Add as many rows as necessary for this function.

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

[Section Head (NCMs) OR HOD (Officers) Title] Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in QSP-AIIW except as noted has successfully completed on the job training. The member is qualified to carry out the duties of Submarine Naval Communicator and is AIIW qualified Naval Communicator 00299 109271.

_____

Divisional Officer Date

ON-JOB PERFORMANCE RECORD SHEET

8. The remarks column in this annex shall be used for the following information only:

a. The unit in which the training and assessment were carried out; and

b. Any unusual circumstances which preclude compliance with an objective.

PO/EO	PERFORMANCE STATEMENT	DATE	SIGNATURE	REMARKS (INCLUDE UNIT)
PO 002	Operate Submarine Specific Communications Equipment			

EO 002.02	Conduct BSCT procedures to include: Configure Satelite Communication BCST			
	Configure HF BCST			
	Configure LF BCST			
EO 002.03	Conduct Ship/Shore procedures to include: Describe Submarine PM/OTO's			
	Tune HF for Submarine Ship/Shore			
	Configure DAMA for Submarine Ship/Shore			
	Configure HF for RATT Ship/Shore Answering			
EO 002.04	Conduct Submarine Ship/Ship Ship/Air Procedures to include: Configure HF for Secure Voice			
	Configure HF for CRATT			
PO/EO	**PERFORMANCE STATEMENT**	**DATE**	**SIGNATURE**	**REMARKS (INCLUDE UNIT)**
	Configure UHF for LOS CRATT			
	Configure HF for P/L Voice			
	Configure UHF for Secure Voice			
	Configure UHF for P/L Voice			
	Configure ECB			
	Configure DAMA for Sec Voice			
	Configure VHF for Multiple/Dual Watch			

PO 005	Operating Submarine Specific Monitoring Equipment			
EO 005.01	Utilizing Atmospheric Monitoring Equipment			
EO 005.02	Disseminating Atmospheric Results			
EO 005.03	Advising Command on Atmospheric Results			

ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
C&A	CERTIFICATION AND ACCREDITATION
CCA	CURRICULUM CONTROL AUTHORITY
CFFSE	CANADIAN FORCES FLEET SCHOOL ESQUIMALT
CFNOS	CANADIAN FORCES NAVAL OPERATIONS SCHOOL
CFSCE	CANADIAN FORCES SCHOOL OF COMMUNICATIONS AND ELECTRONICS
CIK	CRYPTO IGNITION KEY
CISN	COMMUNICATIONS AND INFORMATION SYSTEMS NETWORK
CMIS	CONSOLIDATED MAINTENANCE INFORMATION SYSTEM
COMPUSEC	COMPUTER SECURITY
COMSEC	COMMUNICATION SECURITY
CSNI	CONSOLIDATED SECRET NETWORK INFRASTRUCTURE
DHCP	DYNAMIC HOST CONTROL PROTOCOL
DIN	DEFENCE INFORMATION NETWORK
DP	DEVELOPMENTAL PERIOD
DTD	DATA TRANSFER DEVICE
DWAN	DEFENCE WIDE AREA NETWORK
EC	ENABLING CHECK
EMSEC	EMISSION SECURITY
EO	ENABLING OBJECTIVE

ACRONYM / ABBREVIATION	TERM/DEFINITION
FFV	FIRE FLY VECTOR SET
FTR	FIELD TAMPER RECOVERY
HSDC	HIGH SPEED DATA CONNECTIVITY
IAW	IN ACCORDANCE WITH
IMD	INFORMATION MANAGEMENT DIRECTOR
INFOSEC	INFORMATION SECURITY
IS	INFORMATION SYSTEM
IS	SECUR INFORMATION SYSTEM SECURITY
ITSEC	INFORMATION TECHNOLOGY SECURITY
KMID	KEY MATERIAL IDENTIFICATION
LCMM	LIFE CYCLE MATERIAL MANAGER
MCOIN	MARITIME COMMAND OPERATIONAL INFORMATION NETWORK
N/A	NOT APPLICABLE
NAVIS	NAVAL INFORMATION SYSTEMS
NETSEC	NETWORK SECURITY
NTI	NAVAL TRAINING INSTRUCTION
OS	OPERATING SYSTEM
OST	OUT SOURCE TRAINING
QS & TP	QUALIFICATION STANDARD AND TRAINING PLAN
PC	PERFORMANCE CHECK
PO	PERFORMANCE OBJECTIVE
PRB	PROGRESS REVIEW BOARD
QSP	QUALIFICATION STANDARD AND PLAN
RAID	REDUNDANT ARRAY OF INDEPENDENT DISKS
RFC	REQUEST FOR CHANGE
RFS	REQUEST FOR SERVICE
SMS	SHORT MESSAGE SERVICE
TACLANE	TACTICAL LOCAL AREA NETWORK ENCRYPTION

ACRONYM / ABBREVIATION	TERM/DEFINITION
TCI	TECHNICAL COMSEC INSPECTION
TCP/IP	TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL
TRANSEC	TRANSMISSION SECURITY
TRB	TRAINING REVIEW BOARD
VOIP	VOICE OVER INTERNET PROTOCOL
AAW	ANTI-AIR WARFARE
A/G	ADDRESS GROUPS
AIG	ADDRESS INDICATING GROUP
ASMD	ANTI SHIP MISSILE DEFENCE
C/S	CALL SIGNS
CCI	CONTROLLED CRYPTOGRAPHIC ITEM
CCMS	COMMUNICATIONS CONTROL MANAGEMENT SYSTEM
CGL	COMMAND GUARD LIST
CGS	COMMUNICATION GATEWAY SYSTEM
CISN OP	COMMUNICATION INFORMATION SYSTEMS AND NETWORK OPERATOR
CMTU	COMPUTER MAGNETIC TAPE UNIT
COMMPLANS	COMMUNICATION PLANS
COMMSPOTS	COMMUNICATION SPONTANEOUS
COMMSTATS	COMMUNICATION STATUS
COMPUSEC	COMPUTER SECURITY
COMSEC	COMMUNICATION SECURITY
CPF	CANADIAN PATROL FRIGATE
CRATT	COVERED RADIO TELETYPE
CSRF	COMMON SOURCE ROUTE FILE
C4I	COMMAND, CONTROL, COMMUNICATIONS AND COMPUTER INFORMATION
DAMA	DEMAND ASSIGNED MULTIPLE ACCESS
DIMS	DAILY INTENTIONS MESSAGE SUMMARY
DMTE	DIRECTOR MARITIME TRAINING AND EDUCATION
DSC	DIGITAL SELECTIVE CALLING

ACRONYM / ABBREVIATION	TERM/DEFINITION
DTD	DATA TRANSFER DEVICE
EAW	ELECTRONIC AND ACOUSTIC WARFARE
EC	ENABLING CHECK
ECL	ENABLING CHECK LIST
EMI	ELECTROMAGNETIC INTERFERENCE
EMSEC	EMISSION SECURITY
EO	ENABLING OBJECTIVE
EPIRB	EMERGENCY POSITION INDICTOR RADIO BEACON
FSK	FREQUENCY SHIFT KEYING
GMDSS	GLOBAL MARITIME DISTRESS AND SAFETY SYSTEM
GUI	GRAPHIC USER INTERFACE
HF	HIGH FREQUENCY
HIDR	HANDLING INFORMATION DESTRUCTION RECORD
HVU	HIGH VALUE UNIT
ISSEC	INFORMATION SYSTEM SECURITY
ITSEC	INFORMATION TECHNOLOGY SECURITY
I/S	INTERSHIP
IMM	INTERNATIONAL MARITIME MOBILE
INFOSEC	INFORMATION SECURITY
INTERCO	INTERNATIONAL CODES
IS	INFORMATION SECURITY
I/S	INTER-SHIP
IS SEC	INFORMATION SYSTEM SECURITY
IT	INFORMATION TECHNOLOGY
IT SEC	INFORMATION TECHNOLOGY SECURITY
JBS	JOB BASED SPECIFICATIONS
JRCC	JOINT RESCUE COORDINATION CENTRE
LAN	LOCAL AREA NETWORK
LAR	LOCAL ADDRESS ROUTER
LCR	LOCAL CONTENT ROUTER

ACRONYM / ABBREVIATION	TERM/DEFINITION
LFSR	LOW FREQUENCY SATELLITE RECEIVER
LPI	LOW PROBABILITY OF INTERCEPT
LRPA	LONG RANGE PATROL AIRCRAFT
MARLANT	MARITIME ATLANTIC
MARPAC	MARITIME PACIFIC

ACRONYM / ABBREVIATION	TERM/DEFINITION
MIJI	MEACONING INTERFERENCE JAMMING INTRUSION
MIO	MARITIME INTERDICTION OPERATIONS
MMSS	MAGNETIC MEDIA STORAGE SYSTEM
MOB	MAN OVERBOARD
NATO	NORTH ATLANTIC TREATY ORGANIZATION
NCS	NET CONTROL STATION
NCTAMS	NAVAL COMPUTER AND TELECOMMUNICATIONS AREA MASTER STATION
NOS	NETWORK OPERATING SYSTEM
NTI	NAVAL TRAINING INSTRUCTION
OGD	OTHER GOVERNMENT DEPARTMENTS
OHP	OVER HEAD PROJECTOR
OJPR	ON JOB PERFORMANCE REQUIREMENTS
OPORD	OPERATION/OPERATIONAL ORDER
OPTASK	OPERATIONAL TASKING
OPSUM	OPERATIONAL SUMMARY
OS	OPERATING SYSTEM
OTAD	OVER THE AIR DISTRIBUTION
OTAR	OVER THE AIR RE-KEY/RECEIVE
OTAT	OVER THE AIR TRANSFER/TRANSMIT
OTO	OPERATOR TO OPERATOR
OOW	OFFICER OF THE WATCH
PC	PERFORMANCE CHECK
PCL	PERFORMANCE CHECK LIST
PLA	PLAIN LANGUAGE ADDRESS
PLAD	PLAIN LANGUAGE ADDRESS DATA BASE
PO	PERFORMANCE OBJECTIVE
PSK	PHASE SHIFT KEYING
RACE	RAPID AUTOMATIC CRYPTO EQUIPMENT
RADHAZ	RADIO HAZARD
RAS	REPLENISHMENT AT SEA
RATT	RADIO TELETYPE

ACRONYM / ABBREVIATION	TERM/DEFINITION
RF	RADIO FREQUENCY
RI	ROUTING INDICATOR
ROC	RECORD OF CHANGES
RPTS	REPORTS
RSI	RADIATION STATUS INDICATOR
RTCU	RADIO TRANSMITTER CONTROL UNIT
S/S	SHIP SHORE
SACEUR	SUPREME ALLIED COMMAND EUROPE
SACLANT	SUPREME ALLIED COMMAND ATLANTIC
SAG	SURFACE ACTION GROUP
SAR	SEARCH AND RESCUE
SART	SEARCH AND RESCUE TRANSPONDER
SARSAT	SEARCH AND RESCUE SATELLITE
SATCOM	SATELLITE COMMUNICATION
SAU	SEARCH ATTACK UNIT
SHD	SPECIAL HANDLING DESIGNATIONS
SHINCOM	SHIPS INTERNAL COMMUNICATION
SSR	SATELLITE SIGNAL RECEIVER
SVC	SERVICE
S/S	SHIP SHORE
TACSIGS	TACTICAL SIGNALS
TF	TASK FORCE
TG	TASK GROUP
TU	TASK UNIT
TGOA	TASK GROUP OPERATIONS ADMIN
TGO	TASK GROUP OPERATIONS
TMS	TEST MEASUREMENT SYSTEM
TRANSEC	TRANSMISSIONS SECURITY
TTY	TELETYPE
UHF	ULTRA HIGH FREQUENCY
UISS	UNIT INFORMATION SECURITY OFFICER
URD	UN-RECOGNIZED DATA

ACRONYM / ABBREVIATION	TERM/DEFINITION
USN	UNITED STATES NAVY
VHF	VERY HIGH FREQUENCY
VTMS	VESSEL TRAFFIC MANAGEMENT SYSTEM
WADS	WIDE AUDIO DATA SWITCH
WT	WARNING TERMS

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

Based on Optimal course load

PO/EO	OBJECTIVE	EO	EC	PC	TOTAL PERIODS
EO 001.01	Configure MHDS for Submarine Broadcast	2	6		
EO 001.02	Configure MHDS for Submarine Ship/Shore	2	6		
EO 001.03	Configure MHDS for Submarine Inter Ship	2	6		
EO 001.04	Configure MHDS for System Backup	4.5	12		
PO 001	Configure MHDS			24	64.5
EO 002.01	Identify Equipment Capabilities and Limitations	2.5	1		
EO 002.02	Conduct Broadcast Procedures	8.5	24		
EO 002.03	Conduct Ship/Shore Procedures	7.8	24		
EO 002.04	Conduct Ship/Ship Ship/Air Procedures	16.6	48		
PO 002	Operate Submarine Specific Communications Equipment			OJT	132.4 + OJT
EO 003.01	Disseminate Key Information	3	1		
EO 003.02	Draft Submarine Formatted Messages	3	1		
PO 003	Perform COMCEN Functions			1	9
EO 004.01	Formulate Recommendations	3	2		
EO 004.02	Deliver Reports	3	1		

PO/EO	OBJECTIVE	EO	EC	PC	TOTAL PERIODS
PO 004	Advise Command on Status of Communications			2	11
EO 005.01	Utilize Atmospheric Monitoring Equipment	2	.25		
EO 005.02	Disseminate Atmospheric Results	1	.25		
EO 005.03	Advise Command on Atmospheric Results	1	.25		
PO 005	Operate Submarine Specific Atmospheric Monitoring Equipment			OJT	5.25 + OJT
Course	31 Training Days				222.15

ANNEX H – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

N/A

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

1. QSP writing Board added Configure Submarine MHDS to the QSP because it was not previously taught for a Submarine environment.

ANNEX K – EVALUATION MATRIX

Note: This annex is optional and **may be** deleted.

EC/PC	Scope Covered	Purpose	How	When	Time	Resources	Limitations	Impact of Limitations
EC 001.01	EO 001.01	To evaluate the trainees ability to configure MHDS for Sub BCST	Practical Assessment	O/C of EO 001.01	50 min Per trainee	MHDS Trainer, Imaging Software	Availability of trainer	Unable to train
EC 001.02	EO 001.02	To evaluate the Trainee's ability to Configure MHDS for Sub Ship/Shore	Practical Assessment	O/C of EO 001.02	50 min Per trainee	MHDS Trainer, Imaging Software	Availability of trainer	Unable to train
EC 001.03	EO 001.03	To evaluate the Trainee's ability to Configure MHDS for Sub I/S	Practical Assessment	O/C of EO 001.03	50 min Per trainee	MHDS Trainer, Imaging Software	Availability of trainer	Unable to train
EC 001.04	EO 001.04	To evaluate the trainee's ability to configure MHDS for Sub system back-up	Practical Assessment	O/C of EO 001.04	100 min Per trainee	MHDS Trainer, Imaging Software	Availability of trainer	Unable to train
PC 001	PO 001	To evaluate the trainee's ability to configure MHDS	Practical Assessment	O/C of PO 001	200 min Per trainee	MHDS Trainer, Imaging Software	Availability of trainer	Unable to assess
EC 002.01	EO 002.01	To identify equipment Capabilities and Limitations	Written exam	O/C of EO 002.01	50 min	Trainer	Nil	Nil
EC 002.02	EO 002.02	To evaluate the trainee's ability to Conduct BCST Procedures	Practical/ Written Assessment	O/C of EO 002.02	100 min Per trainee	Trainer or Vic Class Sub	Availability of Trainer/Sub	Unable to assess practical
EC 002.03	EO 002.03	To evaluate the trainees ability to Conduct Ship/Shore procedures	Practical exam	O/C of EO 002.03	200 min Per trainee	trainer or Vic Class Sub	Availability of Trainer/Sub	Unable to assess
EC 002.04	EO 002.04	To evaluate the trainee's ability to conduct Ship/Ship and Ship/Air Procedures	Practical exam	O/C of EO 002.04	400 min per trainee	trainer or Vic Class Sub	Availability of Trainer/Sub	Unable to assess
PC 002	PO 002	To evaluate the Trainee's ability to operate Sub specific communications equipment	Practical Exam	O/C of PO 002	200 min Per trainee	Trainer or Vic Class Sub	Availability of Sub	Unable to Assess
EC 003.01	EO 003.01	To evaluate the trainee's disseminate key information	Written exam	O/C of EO 003.01	50 min	Trainer	Nil	Nil
EC 003.02	EO 003.02	To evaluate trainee's ability to draft Sub specific messages	Written exam	O/C of EO 003.02	50 min	Trainer	Nil	Nil
PC 003	PO 003	To evaluate trainee's ability to perform COMCEN functions	Written exam	O/C of PO 003	50 min	Trainer	Nil	Nil
EC 004.01	EO 004.01	To evaluate trainee's ability to formulate recommendations to command	Oral exam	O/C of EO 004.01	100 min	Trainer	Nil	Nil
EC 004.02	EO 004.02	To evaluate the trainee's ability to deliver reports to command	Written/ Oral exam	O/C of EO 004.02	50 min	Trainer	Nil	Nil
PC 004	PO 004	To evaluate the trainee's ability to advise command on the status of communications	Practical exam	O/C of PO 004	100 min	Trainer or Vic Class Sub	Nil	Nil
EC 005.01	EO 005.01	To evaluate the trainee's ability to utilize atmospheric monitoring	Oral confirmation	O/C of EO005.01	50 min	Trainer	Nil	Nil

		equipment						
EC 005.02	EO 005.02	To evaluate the trainee's ability to disseminate atmospheric results	Oral confirmation	O/C of EO 005.02	50 min	Trainer	Nil	Nil
EC 005.03	EO 005.03	To evaluate the trainee's ability to advise command on atmospheric results	Oral confirmation	O/C of EO 005.03	50 min	Trainer	Nil	Nil
PC 005	PO 005	To evaluate the trainee's ability to operate Sub specific atmospheric monitoring equipment	Practical Exam	On OJT	50 min	Trainer or Vic Class Sub	Availability of Sub	Unable to Assess